

Competition and Pricing Dynamics in the UK Espresso Machine Digital Market

1. Data Preparation

2. Market Concentration

3. Empirical Analysis

4. Synthesis and Reflection

5. Technical Appendix

6. Bibliography

Data Preparation

The empirical foundation utilises a primary dataset extracted from Amazon.co.uk on the 7th March 2026. Targeting the espresso machine category simulated a representative consumer search journey. A VBA script initially acquired 309 unique observations identified by Amazon Standard Identification Numbers (ASIN), including both organic and algorithmically injected sponsored advertisements. Scraping all seven result pages captured the active digital shelf, ensuring no significant brands or variants were omitted. Filtering removed duplicates and unpriced observations to focus on organic competition. This resulted in a refined baseline of 303 observations for the market universe.

To facilitate a comprehensive competition analysis, the dataset includes product characteristics beyond pricing. Using an invoked function within Power Query, the dataset was expanded to include brand, pump pressure, capacity, weight, review scores, review count, and special features. Technical specifications extracted from these features were used to create dummy variables regarding the presence of integrated grinders, milk-frothing systems, and manual operation. This breadth of data allows for the quantification of quality dimensions and technological features that directly influence consumer willingness to pay. Following the automated scrape and the removal of duplicates, the dataset comprised 303 observations, which provided a comprehensive foundation for the cleaning and enrichment phase.

The raw metadata underwent transformation to ensure suitability for econometric modelling. Observations lacking brand names or prices were removed to preserve integrity. The pump pressure variable suffered from near-total missingness, which necessitated its removal. Capacity and weight specifications were cleaned by stripping non-numeric strings and standardising values into litres and kilograms, respectively. A manual outlier audit corrected unit anomalies and refined the market definition by removing non-espresso machines. To address a 30% completion rate for weight, a significance-weighted enrichment strategy used manual research to recover data for market-leading brands. This yielded 183 observations for concentration analysis and a sub-sample of 111 observations for controlled regression analysis.

Market Concentration

2.1 Manufacturers

The espresso machine market on Amazon UK comprises established global multinationals and numerous smaller generic entrants. Primary players identified in the dataset include De'Longhi, Nespresso, and Krups, which are prominent multinational corporations with extensive manufacturing networks. De'Longhi operates as a major Italian group while Nespresso and Krups are part of the Swiss-based Nestlé and French-based Groupe SEB, respectively. These firms leverage significant economies of scale and brand equity to maintain their positions. The market also includes specialised firms such as Sage which is part of the Australian Breville Group. While these multinationals dominate premium segments, the digital shelf is populated by generic or domestic firms focusing on entry level machines. This suggests that while high end engineering requires multinational capital, the barriers to entry for basic pump machines are low enough to allow smaller firms to compete on price.

2.2 Concentration Analysis

Market shares are calculated using product listings to represent variety and consumer reviews to proxy for sales volume. These measures assume functional substitutability within the defined market and that manufacturers maintain relatively symmetric cost structures for comparable technology. The competitive landscape is quantified through the Four Firm Concentration Ratio (C4) and the Herfindahl Hirschman Index (HHI).

The C4 ratio is defined as the sum of the market shares of the four largest firms in the industry. This ratio approaches 0% in a perfectly competitive market and reaches a maximum of 100% in a market entirely controlled by four entities.

HHI is defined by the equation,

$$HHI = \sum_{i=1}^N (S_i)^2,$$

where the percentage market share S_i of each individual firm is squared and summed for all N firms in the market.

This index ranges from a theoretical minimum of $\frac{10,000}{N}$ to a maximum value of 10,000. With 183 firms identified in the current market the theoretical minimum HHI is 54.64.

Table 1: Market Concentration Comparison

Ratio	Count of product listings	Count of reviews
C4 Ratio	18.58%	70.20%
HHI	197.38	1785.54

Results in Table 1 demonstrate that the count of product listings indicates a highly competitive environment, whereas the count of reviews reveals a significantly higher degree of concentration. This disparity suggests that although hundreds of brands are visible to consumers, the actual purchasing activity is concentrated within a small elite group of trusted manufacturers.

2.3 Merger Analysis

The concentration of the market is further quantified using two distinct hypothetical merger scenarios between the firm with the highest market share and the firm with the third-largest share. For the product listing measure this merger was analysed between De'Longhi and Krups. For the reviews measure the merger was analysed between De'Longhi and Nespresso.

Table 2: Impact of Hypothetical Mergers on concentration measures

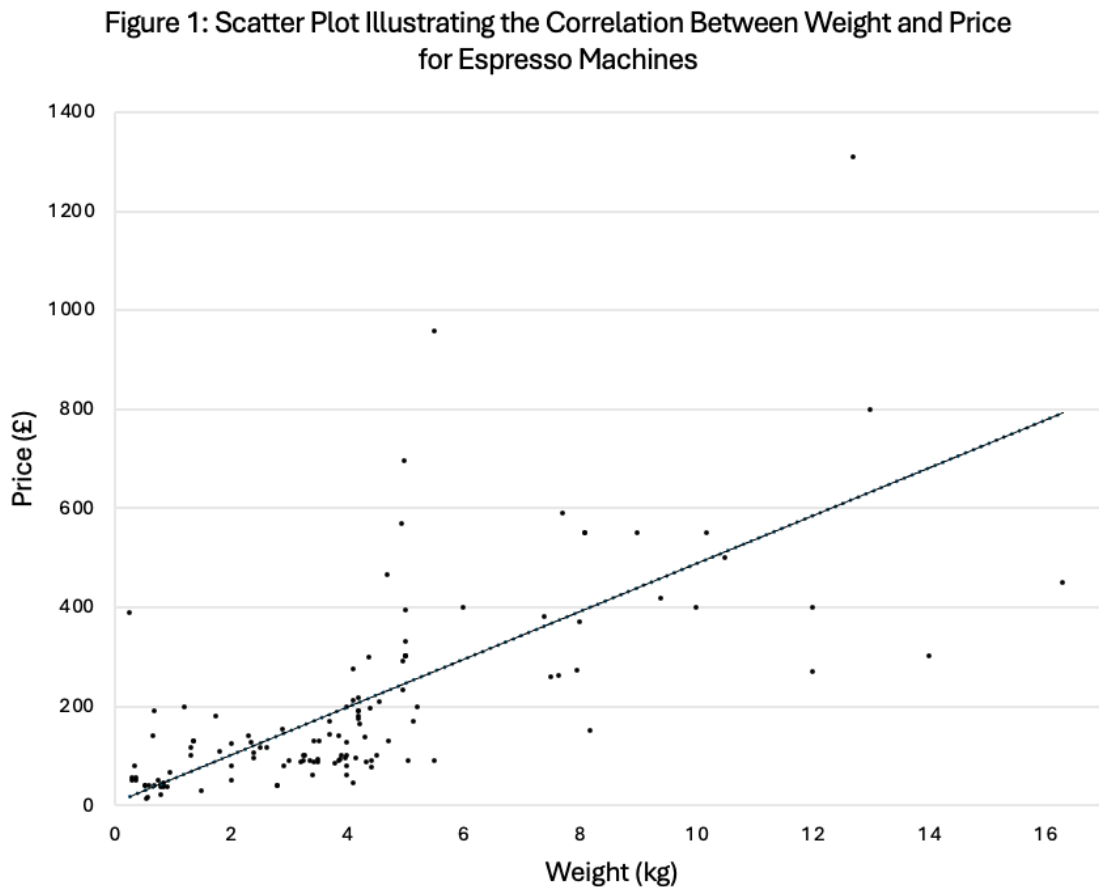
Ratio	Product Listings with merger between De'Longhi and Krups	Reviews with merger between De'Longhi and Nespresso
C4 Ratio	21.86%	75.28%
HHI	251.72	2382.99

The results in Table 2 show that the merger between De'Longhi and Nespresso would cause the reviews HHI to increase by 597.45 to reach a total of 2382.99. This represents a significant shift that moves the market structure closer to the threshold of high concentration. Such an increase would likely trigger regulatory scrutiny from the Competition and Markets Authority because it represents a substantial reduction in competitive pressure within a moderately concentrated market. This merger would grant the combined entity an overwhelming level of market engagement which could lead to higher prices and a reduction in innovation across the UK espresso machine segment.

Empirical Analysis

3.1 Preliminary Analysis

To establish the empirical foundation for the hedonic models, a bivariate analysis was conducted to explore the relationship between weight and market value. Figure 1 illustrates the correlation, providing a visual justification for including weight as a primary independent variable.



3.2 Empirical Results

Following the preliminary visual analysis, regression specifications were estimated. Table 3 presents the comparative results.

Table 3: Comparative Hedonic Price Analysis

Model	Naïve (weight)	Naïve (manual)	Linear	Non-linear
Dependent variable	Price (£)	Price (£)	Price (£)	Log of Price (£)
Variables				
Weight (kg)	48.33*** (8.664)		55.99*** (14.06)	0.138*** (0.0345)
Manual		197.0 (145.6)	196.6 (157.0)	0.257 (0.212)
Count of reviews			-0.00167 (0.00128)	-0.00000640 (0.00000561)
Capacity (litres)			-84.24 (78.11)	-0.0149 (0.199)
Reviews (out of 5)			-53.78 (48.58)	-0.0514 (0.0787)
Milk Frother			31.07 (59.21)	0.191 (0.193)
Grinder			2.809 (58.71)	-0.180 (0.234)
Brand	Not Included	Not Included	Included	Included
Constant	6.123 (27.64)	186.0*** (14.87)	210.8 (243.9)	4.291*** (0.454)
R^2	0.464	0.041	0.888	0.929
Observations	125	183	111	111

Robust standard errors reported in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Full variable definitions are provided in Appendix 5.1.

3.3 Baseline Specification

The analysis begins with two naïve specifications to establish baseline correlations between price and individual product characteristics. As presented in Table 3, an initial naïve regression focusing solely on machine weight explains 46.4 per cent of price variation. A secondary baseline model examines manual functionality, which shows a significantly weaker initial correlation of 4.1 per cent. These specifications provide a necessary contrast to the full hedonic pricing model, which posits that a product is a bundle of characteristics and its price is the sum of the valuations of those individual attributes.

The linear specification can be defined as,

$$P_i = \beta_0 Weight_i + \beta_1 Manual_i + X_i' \beta + \epsilon_i.$$

In this model, P_i represents the price for product i and $X_i' \beta$ is a vector of control variables that incorporates technical features such as capacity, grinder presence, and milk frothing capabilities, alongside categorical brand identifiers. The composite error term ϵ_i contains the constant intercept and the idiosyncratic error term.

Consequently, the full linear specification is justified as it moves the variables within the $X_i' \beta$ vector out of the error term and into the regression model. This addresses the omitted variable bias present in the baseline models, where the exclusion of brand equity and technical specifications caused the regressors to be correlated with the error term. This consolidated approach isolates the impact of tangible engineering from manufacturer prestige, successfully increasing the explanatory power to 88.8 per cent.

3.4 Diagnostics

The full linear regression suggests that weight is the most robust factor of price. *Ceteris paribus*, each additional kilogram of weight adds £55.99 to the predicted retail price ($p < 0.01$). This reflects a high willingness to pay for build quality. Interestingly, while manual functionality and milk frothers carry positive coefficients, they are not statistically significant in this linear specification when controlling for weight and brand.

The diagnostic phase identified significant heteroscedasticity through a Breusch-Pagan test ($p = 0.000$), confirming that error variance was not constant across the price range. This was addressed by using robust standard errors for all estimates to ensure the reliability of t-statistics. Additionally, a Ramsey RESET test was conducted ($p = 0.000$), indicating that the linear functional form was misspecified.

3.5 Non-Linear Specification

Following the diagnostic evidence from the RESET test, the model was re-estimated using the natural logarithm of price as the dependent variable. This non-linear specification captures 92.9% of price variation and allows for the interpretation of coefficients as percentage effects.

In the non-linear model, the coefficient of 0.138 indicates that a one-kilogram increase in weight is associated with an approximately 13.8% increase in price ($p < 0.01$). While Manual products suggest a 25.7% premium, it remains statistically insignificant, reinforcing the finding that in this Amazon-based market, physical build quality and brand identity are more consistent predictors of price.

3.6 Market Context

The regression findings align with the concept of vertical product differentiation, where firms use high quality physical engineering and reputation to distance themselves from competitors in an oligopolistic landscape (Fiorani, 2017). These results are supported by the analysis of the international espresso market, which identifies that technical product quality and brand perception are primary drivers of market value (Fiorani, 2017). The massive explanatory power of the weight variable and brand controls suggests that manufacturers create competitive advantages through build quality, effectively protecting market share against generic competitors (SEB and WMF, 2016).

This statistical evidence is further validated by regulatory decisions and cultural analyses of the coffee sector. The European Commission (2016) investigation into the SEB and WMF merger highlighted a distinct prosumer segment where consumers prioritise professional grade technical specifications for domestic use. This aligns with the cultural shift identified by Morris (2013), where the move toward espresso signifies a desire to replicate the professional coffee house experience at home through specialised equipment. These sources provide a real world justification for the high valuation of both build quality and brand identifiers found in the model.

Furthermore, the necessity of a log linear specification to achieve a superior statistical fit aligns with observations in the Arçelik and Whirlpool (2024) report. The Competition and Markets Authority noted that technical innovations and brand equity often add proportional, percentage-based value to appliances rather than fixed marginal costs, justifying the percentage premium found in the non-linear model. While the market faces constraints from new entrants and private labels, the robust willingness to pay for machine weight suggests that in the premium espresso segment, technical differentiation remains a more potent price determinant than pure cost leadership strategies (SEB and WMF, 2016).

Synthesis and Reflection

4.1 Conclusion

Analysis of the Amazon UK marketplace identifies a vertically differentiated oligopoly characterised by high concentration among multinational incumbents. Structural findings from Section 3 confirm that a limited number of firms maintain significant market power, while empirical results in Section 4 identify physical weight as a statistically significant determinant of price. These results indicate that competition is driven by technical differentiation and brand reputation, as the platform ecosystem reinforces established brand authority.

The structural analysis in Section 3 acknowledges that review volume, while a standard proxy for market power, remains constrained by brand-specific review-to-sale variance. Furthermore, the robustness of the econometric findings is limited by the OLS exogeneity assumption. This requires the error term ϵ_i to be uncorrelated with the regressors X . If this condition fails such that $Cov(X, \epsilon_i) \neq 0$, the resulting coefficients become biased and inconsistent. To mitigate cross-sectional constraints, future research should leverage longitudinal panel data to evaluate the temporal stability of engineering-led price premiums. Firm-level revenue data would permit a decomposition of market power, distinguishing between technical innovation and brand equity to facilitate a more robust competition analysis.

4.2 Theoretical Framework

In accordance with the theoretical frameworks established by Belleflamme and Peitz (2021), the empirical analysis in this report applies models of strategic interaction and platform competition. The log-linear transition in Table 3 aligns with hedonic pricing theories where attributes yield proportional utility increments in premium durable goods markets, rather than fixed marginal increases. The evidence of an uncompetitive market structure, which was characterised by high concentration levels where the C4 index reached $\approx 70\%$ based on review volume, aligns with theories regarding market power and the role of reputation as a barrier to entry. These reputational advantages serve as a mechanism to reduce search frictions for established firms, which effectively heightens the information asymmetry faced by newer entrants. Furthermore, the application of hedonic pricing to isolate tangible attributes like weight and capacity mirrors the econometric approach to resolving omitted variable bias. By synthesising these results with established theories on entry barriers and ecosystem compatibility, this report demonstrates how technical differentiation and platform reputation interact to dictate competitive outcomes in modern marketplaces.

Technical Appendix

5.1 Variable Definitions

Variable	Definition	Data Type
Price	Value of the product In pounds	Numeric
Log of price	The natural log of Price	Numeric
Weight	Weight of the product in kilograms	Numeric
Brand	Manufacturer of the product	Categorical
Capacity	Water tank volume of the product	Numeric
Review score	Mean consumer rating	Numeric
Count of Reviews	Total number of user reviews	Numeric
Manual	1 if the machine is manual	Binary
Milk Frother	1 if an integrated milk frothing system is included	Binary
Grinder	1 if an integrated grinder is included	Binary

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